

Mapping the Milky Way with SALSA

RRY125 Modern Astrophysics– SALSA Lab

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Image credit: NASA/JPL-Caltech [\[1\]](#)

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Abstract

In this report, we measure Galactic neutral hydrogen (HI) using the SALSA telescope to derive the Milky Way’s rotation curve and create a basic face-on map of its HI distribution. Using the 21 cm line and the Doppler effect, measured radial velocities (V_r) were related to distances and constructed a rotation curve, which agrees with the established nearly flat galacto-centric radii. Additionally, the spatial distribution of observed HI clouds was plotted by identifying spiral arms. Distance ambiguities in certain directions are noted, and potential resolution methods are discussed; however, these solutions are not applied in this mapping of the Milky Way.

1 Introduction

Understanding the Milky Way’s structure and kinetics is important in astrophysics. One key diagnostic is the galactic rotation curve $\mathbf{V}(\mathbf{R})$, which shows how orbital velocity varies with galacto-centric radius $\mathbf{R}$.

Neutral hydrogen (HI) gas is a key component in constructing the Galactic rotation curve and mapping the large-scale structure of the Milky Way. Neutral hydrogen is abundant and can be detected via its spectral line emission at a wavelength of $\lambda = 21$ cm (1420 MHz). This line arises from a spin-flip transition – the electron’s spin flips relative to the proton’s flip. Although this is rare, the massive amount of HI gas in the Galactic Disk produces a detectable radio signal. [2]

The line-of-sight velocities of HI clouds can be determined by measuring the Doppler shifts of the 21 cm line. The Doppler effect enables us to convert frequency shifts into velocities. If the gas moves towards us, the line is shifted to higher frequencies; if it moves away, it shifts to lower frequency. We express this quantitatively with the Doppler equation:

$$\frac{f - f_0}{f_0} = \frac{v}{c} \quad (1)$$

- $\Delta f = f - f_0$: frequency shift
- f : observed frequency,
- f_0 : observed rest frequency of the line
- v : velocity:
 - $v > 0$ if the object is receding,
 - $v < 0$ if it is approaching.

This relation allows us to calculate the velocities of hydrogen clouds from their frequency shifts. Coupled with knowledge of the Sun’s orbital velocity and position in the Milky Way, these velocity measurements allow us to infer galacto-centric distances for the observed clouds, construct a rotation curve, and map the spatial distribution of HI gas.

In this project, it is used the SALSA (Such A Lovely Small Antenna) telescope at the Onsala Space Observatory [3]. The SALSA telescope is a 2.3 m radio dish sensitive enough to detect HI emission from distant parts of the Galaxy.

2 Data Acquisition and Processing

After obtaining the raw spectra from the SALSA telescope, each spectrum was post-processed to improve data quality. First, a baseline subtraction was performed to remove any continuum background and instrumental offsets. Next, Gaussian functions were fitted to the spectral peaks corresponding to HI emission, allowing for a more precise determination of the peak velocities. These steps were carried out using MATLAB, utilizing libraries and example code provided by SALSA with MATLAB.[4]

3 Constants and Parameters

The standard Galactic constants for the Milky Way are as follows:

- $R_0 = 8.5$ kpc: The distance of the Sun from the Galactic center.
- $V_0 = 220$ km/s: The orbital velocity of the Sun around the Galactic center.

These values are commonly referenced in studies of Galactic dynamics.

4 Constructing the Rotation Curve

4.1 Methods

To construct the Galactic rotation curve, the focus was on Galactic longitudes ℓ in the range from 18° up to about 110° , observing at intervals of a few degrees (e.g., $\ell = 18, 20, 22, 25, 27, \dots, 110$). The set of observed directions is illustrated conceptually in Figure 2. From each direction, a spectrum was obtained, and the maximum radial velocity component, $v_{r,\max}$, corresponding to the tangent point, was identified.

At the tangent point, the geometry of the Galactic disk allows us to simplify the relationship between observed velocities and Galactic parameters. From the SALSA project documentation [5], the relevant equations are:

$$R = R_0 \sin(\ell) \quad (2)$$

$$V(R) = v_{r,\max} + V_0 \sin(\ell) \quad (3)$$

In Figure 1, we see the geometric configuration: the line-of-sight SM touches the tangent point where the maximum projected velocity is observed. By fitting Gaussians to the spectral peaks, we obtained more accurate $v_{r,\max}$ values. Applying the above equations for each observed longitude gave us a set of $(R, V(R))$ points.

4.2 Results

After processing the data and applying the equations, we derived a rotation curve for the Milky Way. The resulting curve (Figure 3) is presented alongside reference rotation curves from external galaxies (Figure 4) for comparison. A full list of the observed longitudes, measured velocities, and calculated $(R, V(R))$ pairs is provided in Appendix A.

As illustrated in Figure 3, the rotational velocity rises rapidly as we move away from the Galactic center, then remains relatively stable at larger R . This pattern contrasts with the Keplerian decline expected if mass were concentrated solely in visible components. Instead, beyond a few kiloparsecs from the center, the rotation curve remains nearly flat, suggesting the presence of significant, unseen mass—commonly attributed to dark matter.

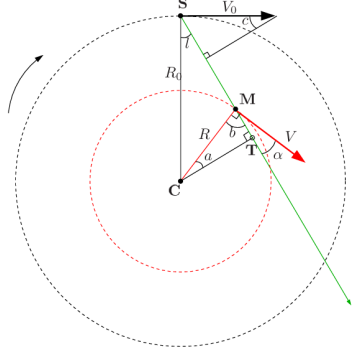


Figure 1: Geometry of the Galaxy. C is the location of the Galactic center, S is the Sun, and M is a gas cloud along the line-of-sight SM. Arrows indicate the direction of rotation and velocities (V_0 for the Sun and V for the cloud). From [5].

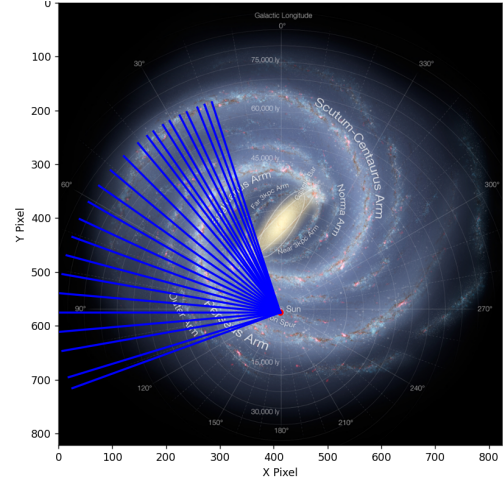


Figure 2: The directions used to construct the rotation curve are shown in blue. Background image from [5]

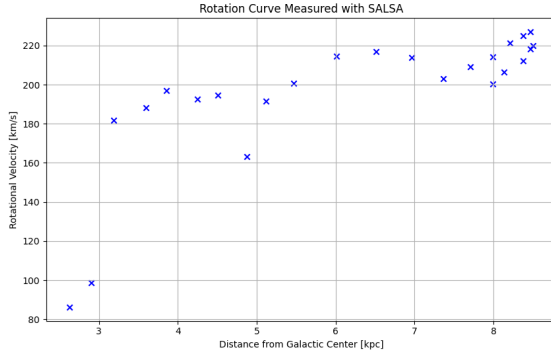


Figure 3: The rotation curve obtained from our observations, showing how $V(R)$ varies with galactocentric radius R .

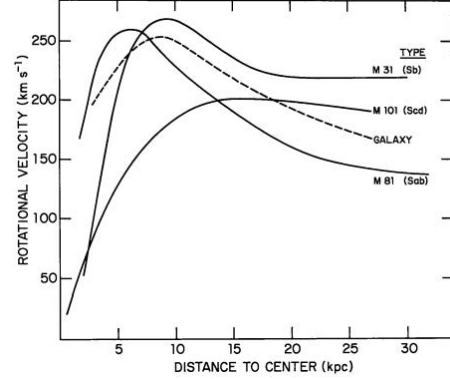


Figure 4: Rotation curves for M31, M101, and M81 obtained by Roberts & Rots (1973) [6], shown for comparison.

For instance, at $R \approx 2.6$ kpc, corresponding to $\ell = 18^\circ$, we find $V(R) \approx 86.3$ km/s. This point lies near the bulge region and confirms the initial rise in velocity. As we extend to larger radii, the velocities do not diminish as would be predicted by a simple Newtonian (Keplerian) drop-off, but stay roughly constant, paralleling the well-studied curves in external galaxies shown in Figure 4.

These results, which can be verified in detail using the tabulated data in Appendix A, align with established Galactic models and serve as a strong validation of our observational technique and chosen constants.

5 Map of the Milky Way

5.1 Methods

To create a face-on map of the Milky Way’s HI distribution, we extended observations beyond the initial range used for the rotation curve. In addition to longitudes up to $\ell = 110^\circ$, directions from $\ell \approx 110^\circ$ to $\ell = 190^\circ$ were included. For each of these observed longitudes (e.g., $\ell = 114^\circ, 118^\circ, 120^\circ, 122^\circ, 126^\circ, 130^\circ, \dots$), significant velocity peaks in the HI spectra were identified. MATLAB was employed to fit Gaussians to these peaks, ensuring accurate determinations of radial velocities.

The Galacto-centric radius R for each velocity component is determined using the equation:

$$R = \frac{R_0 V_0 \sin(\ell)}{V_0 \sin(\ell) + v_r}$$

Once R is known, the line-of-sight distance $r_\pm$ from the Sun is computed using:

$$r_\pm = R_0 \cos(\ell) \pm \sqrt{R^2 - (R_0 \sin(\ell))^2}$$

In the first and fourth Galactic quadrants, this equation can yield two positive solutions (near and far), introducing the so-called distance ambiguity. The correct solution is typically determined by comparing observations at different Galactic latitudes ($b \neq 0^\circ$). If a cloud’s intensity diminishes when observed slightly out of the plane, it suggests the cloud is at the farther distance. Conversely, if the cloud remains bright, it is likely nearer.

For simplicity in this analysis, the distance ambiguity is ignored, and a single solution is assumed. While this approximation may place some clouds incorrectly, it simplifies the analysis. Additional steps to resolve ambiguities are discussed in Section 6.

After determining r , the coordinates are converted to Cartesian coordinates (x, y) in a face-on view centered on the Galactic center. The angle $\theta = \ell - 90^\circ$ is used, with the following equations:

$$x = r \cos(\theta), \quad y = r \sin(\theta).$$

This transformation places the Galactic center at the origin and provides a top-down map of the HI clouds.

5.2 Results

Figure 5 shows a preliminary face-on map of the Milky Way’s HI distribution derived from the measurements. The details of all recorded measurements, including the corresponding (x, y) coordinates, are listed in Appendix B.

Some data points may not align perfectly with expected arm structures. These discrepancies may arise from unresolved distance ambiguities, observational uncertainties, or simplifications in the analysis.

Nonetheless, this preliminary effort demonstrates how HI velocity measurements can be translated into a coherent, large-scale map of the Galaxy’s HI distribution.

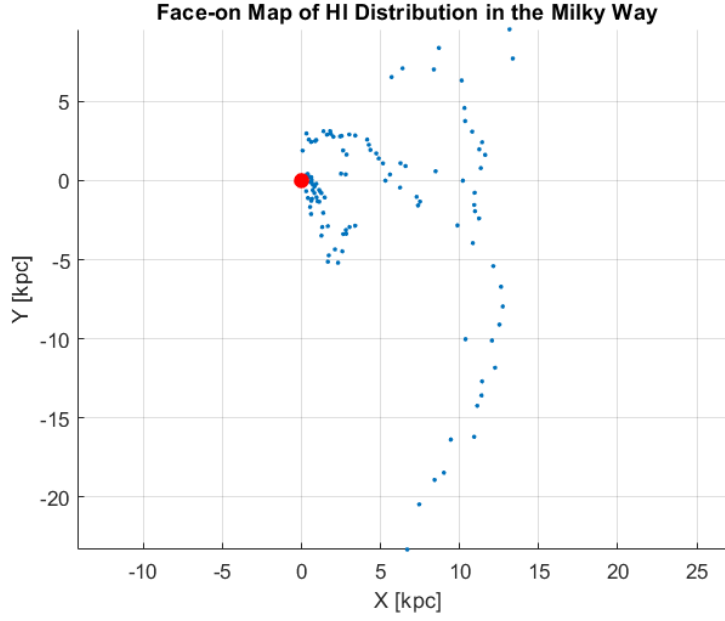


Figure 5: Face-on map of HI distributions. Each point represents an HI cloud, placed according to its derived (x, y) coordinates relative to the Galactic center.

6 Distance Ambiguity

We present the line-of-sight distances for clouds at $b = 0^\circ$ and $\ell = 45^\circ$. The table shows the velocity shift V , the derived Galactocentric radius R , and the two possible line-of-sight distances r_+ and r_- .

Table 1: Line-of-sight distances for clouds at $b = 0^\circ$, $\ell = 45^\circ$

V [km/s]	R [kpc]	r_+ [kpc]	r_- [kpc]	Ambiguity
-54.848	13.129	17.683	-5.662	Unambiguous
6.98658	8.135	11.492	0.529	Ambiguous
22.4452	7.428	10.375	1.645	Ambiguous
57.4848	6.207	7.558	4.463	Ambiguous

- The negative velocity of $V = -54.848$ km/s provides a clear unambiguous positive solution for r_+ .
- For positive velocities ($V = 6.98658$, 22.4452 , and 57.4848 km/s), both potential distances are positive, indicating ambiguity without additional data such as intensity variations at different latitudes.

By observing the same longitude at $b = 4^\circ$, we compared how the intensity of these velocity components changed:

- Velocity $-54.848 \text{ km s}^{-1}$: The corresponding cloud nearly disappears when observing at $b = 4^\circ$. This suggests the cloud is likely at the farther distance.

- Velocity $6.98658 \text{ km s}^{-1}$: The cloud's intensity decreases only slightly at $b = 4^\circ$. This indicates the cloud may be nearer, as it remains detectable even off the plane.
- Velocity $22.4452 \text{ km s}^{-1}$: The intensity decreases a little bit $b = 4^\circ$, suggesting this cloud is likely farther away than the peak at Velocity $6.98658 \text{ km s}^{-1}$ but still near.
- Velocity $57.4848 \text{ km s}^{-1}$: This peak effectively disappears at the higher latitude, indicating a far-distance solution.

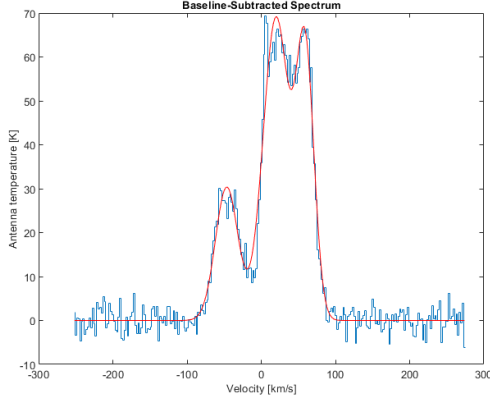


Figure 6: Galactic longitude $\ell = 42^\circ$, latitude $b = 0^\circ$.

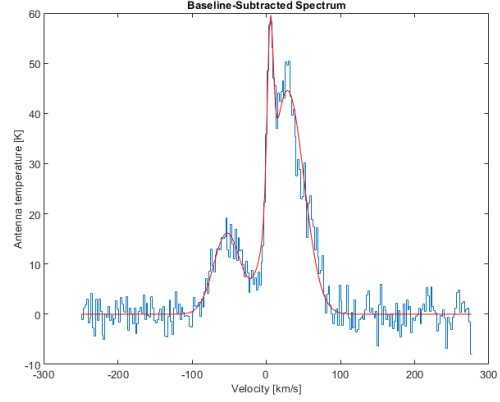


Figure 7: Galactic longitude $\ell = 42^\circ$, latitude $b = 5^\circ$.

By comparing cloud intensities at different latitudes, we identify which solution (near or far) is correct. Clouds that fade drastically at $b = 4^\circ$ are interpreted as far, while those that remain visible are considered near. In this way, observing at multiple latitudes helps break the near-far degeneracy, improving the accuracy of our distance determinations and, consequently, the fidelity of our Galactic HI distribution maps.

References

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- [5] Cathy Horellou, Daniel Johansson, and Eskil Varenus. *SALSA Project Documentation: Mapping the Milky Way*. Revised 2015-12-09 11:53. SE-439 92 Onsala, Sweden, 2015.
- [6] M. S. Roberts and A. H. Rots. “Comparison of Rotation Curves of Different Galaxy Types”. In: *Astronomy and Astrophysics* 26 (1973), pp. 483–485. URL: <https://ui.adsabs.harvard.edu/abs/1973A&A....26..483R>.

A Measurements Data: Rotation Curve

Measurements were taken on **2024-12-18**. Observations were conducted at a frequency of **1420.4 MHz**, using the frequency switching mode with a reference frequency of **1424.4 MHz**. The setup included the following parameters:

- Number of channels: **256**
- Bandwidth: **2.5 MHz**
- Integration time: **20 s**

Longitude (°)	Latitude (°)	Vmax (km/s)	R (kpc)	Vmax
18	0	86.28	2.63	18.30
20	0	98.67	2.91	23.43
22	0	181.83	3.18	99.42
25	0	188.34	3.59	95.36
27	0	196.93	3.86	97.05
30	0	192.71	4.25	82.71
32	0	194.68	4.50	78.10
35	0	163.19	4.88	37.01
37	0	191.44	5.12	59.04
40	0	200.83	5.46	59.42
45	0	214.60	6.01	59.04
50	0	216.91	6.51	48.38
55	0	213.74	6.96	33.53
60	0	202.96	7.36	12.44
65	0	209.29	7.70	9.90
70	0	214.05	7.99	7.32
75	0	221.36	8.21	8.86
80	0	224.98	8.37	8.32
85	0	226.94	8.47	7.78
90	0	220.03	8.50	0.03
95	0	218.11	8.47	-1.05
100	0	212.02	8.37	-4.64
107	0	206.52	8.13	-3.87
110	0	200.49	7.99	-6.25

B Measurements Data: Map of the Milky Way

Measurements were taken on **2024-12-18** using the SALSA telescope at a frequency of **1420.4 MHz**, employing the frequency switching mode. The setup included the following parameters:

- Number of channels: **256**
- Bandwidth: **2.5 MHz**
- Integration time: **20 s**

The following table summarizes the measured data, including Galactic longitudes (L), latitudes (b), velocity shifts (v_{shift}), and derived positions in Galactic coordinates.

L (°)	b (°)	v_{shift} (km/s)	R (kpc)	r (kpc)	x (kpc)	y (kpc)
16	0	19.68	6.42	2.20	0.61	-2.11
16	0	-28.92	16.25	24.25	6.68	-23.31
18	0	85.54	3.76	5.39	1.67	-5.12
18	0	16.27	6.86	1.75	0.54	-1.66
20	0	48.12	5.18	3.69	1.26	-3.47
20	0	11.07	7.41	1.17	0.40	-1.10
20	0	-29.84	14.09	21.77	7.45	-20.46
20	0	-229.45	-4.15	5.03	1.72	-4.73
24	0	98.93	4.04	5.68	2.31	-5.19
24	0	43.65	5.71	3.22	1.31	-2.94
24	0	7.57	7.84	0.73	0.30	-0.67
24	0	-32.67	13.39	20.70	8.42	-18.91
26	0	79.23	4.67	4.83	2.12	-4.34
26	0	16.61	7.25	1.42	0.62	-1.28
26	0	-35.39	13.43	20.54	9.00	-18.46
30	0	85.28	4.79	5.16	2.58	-4.47
30	0	49.49	5.86	3.32	1.66	-2.88
30	0	16.59	7.39	1.32	0.66	-1.14
30	0	-33.92	12.29	18.89	9.45	-16.36
34	0	35.23	6.61	2.46	1.37	-2.04
34	0	-44.73	13.36	19.53	10.92	-16.19
38	0	64.39	5.76	4.29	2.64	-3.38
38	0	22.93	7.27	1.65	1.02	-1.30
38	0	-43.38	12.51	18.06	11.12	-14.23
40	0	63.70	5.86	4.39	2.82	-3.36
40	0	24.36	7.25	1.74	1.12	-1.34
40	0	-44.97	12.46	17.71	11.39	-13.57
42	0	58.96	6.07	4.20	2.81	-3.12
42	0	-44.36	12.17	17.07	11.42	-12.69
42	0	19.48	7.51	1.42	0.95	-1.05

L (°)	b (°)	v_{shift} (km/s)	R (kpc)	r (kpc)	x (kpc)	y (kpc)
46	0	53.82	6.34	4.22	3.03	-2.93
46	0	15.03	7.76	1.12	0.81	-0.78
46	0	-29.94	10.48	14.42	10.37	-10.02
46	0	-52.21	12.68	17.02	12.24	-11.82
50	0	48.76	6.59	4.43	3.40	-2.85
50	0	11.99	7.94	0.93	0.71	-0.60
50	0	-50.61	12.15	15.72	12.04	-10.10
54	0	21.60	7.58	1.81	1.46	-1.06
54	0	-57.27	12.53	15.47	12.52	-9.10
58	0	16.31	7.82	1.48	1.26	-0.79
58	0	-62.05	12.74	15.00	12.72	-7.95
60	0	14.07	7.92	1.34	1.16	-0.67
62	0	12.51	7.99	1.26	1.11	-0.59
62	0	-64.66	12.74	14.29	12.61	-6.71
66	0	8.02	8.17	0.91	0.83	-0.37
66	0	-64.53	12.52	13.28	12.13	-5.40
70	0	5.78	8.27	0.77	0.72	-0.26
70	0	-57.24	11.75	11.53	10.84	-3.94
74	0	4.26	8.33	0.71	0.68	-0.20
74	0	-53.53	11.38	10.26	9.87	-2.83
78	0	3.77	8.35	0.96	0.94	-0.20
78	0	-34.42	10.12	7.53	7.37	-1.57
78	0	-72.11	12.78	11.48	11.23	-2.39
80	0	2.12	8.42	0.59	0.58	-0.10
80	0	-39.17	10.38	7.61	7.49	-1.32
80	0	-72.61	12.78	11.14	10.97	-1.93
82	0	0.58	8.48	0.17	0.17	-0.02
82	0	-40.41	10.44	7.35	7.28	-1.02
82	0	-74.92	12.96	11.03	10.92	-1.54
86	0	-36.44	10.19	6.25	6.23	-0.44
86	0	-80.29	13.40	10.97	10.95	-0.77
90	0	-33.40	10.02	5.31	5.31	0.00
90	0	-79.19	13.28	10.20	10.20	0.00
94	0	-1.76	8.57	0.64	0.64	0.04
94	0	-41.89	10.51	5.61	5.60	0.39
94	0	-69.44	12.43	8.50	8.48	0.59
94	0	-92.22	14.66	11.37	11.34	0.79
98	0	-0.14	8.51	0.04	0.04	0.01
98	0	-19.23	9.32	2.83	2.80	0.39
98	0	-56.71	11.49	6.64	6.58	0.92
98	0	-97.73	15.41	11.73	11.62	1.63
100	0	-18.27	9.28	2.54	2.50	0.44
100	0	-55.96	11.46	6.35	6.25	1.10
100	0	-96.86	15.37	11.42	11.24	1.98
102	0	-1.65	8.57	0.29	0.29	0.06
102	0	-47.21	10.89	5.26	5.15	1.09
102	0	-99.47	15.81	11.68	11.42	2.43
106	0	-2.33	8.59	0.32	0.31	0.09
106	0	-48.65	11.04	5.08	4.88	1.40
106	0	-98.02	15.84	11.23	10.80	3.10
110	0	-5.82	8.75	0.66	0.62	0.22
110	0	-50.68	11.26	5.03	4.73	1.72
110	0	-97.23	16.05	11.01	10.35	3.77

L (°)	b (°)	v_{shift} (km/s)	R (kpc)	r (kpc)	x (kpc)	y (kpc)
114	0	-5.45	8.74	0.55	0.50	0.22
114	0	-49.99	11.31	4.77	4.36	1.94
114	0	-98.42	16.66	11.28	10.30	4.59
118	0	-5.35	8.74	0.49	0.43	0.23
118	0	-51.61	11.58	4.82	4.26	2.26
120	0	-36.83	10.54	3.29	2.85	1.64
120	0	-113.45	21.01	15.43	13.36	7.72
122	0	-6.23	8.79	0.53	0.45	0.28
122	0	-52.69	11.85	4.90	4.15	2.59
122	0	-98.27	17.96	11.95	10.13	6.33
126	0	-37.13	10.74	3.25	2.63	1.91
126	0	-110.28	22.34	16.26	13.16	9.56
130	0	-6.15	8.82	0.49	0.37	0.31
130	0	-47.64	11.85	4.44	3.40	2.85
130	0	-87.30	17.64	10.93	8.37	7.02
134	0	-5.11	8.78	0.40	0.29	0.28
134	0	-44.34	11.81	4.20	3.02	2.92
134	0	-87.41	18.99	12.07	8.68	8.39
138	0	-6.19	8.87	0.49	0.33	0.37
138	0	-39.36	11.60	3.80	2.54	2.82
138	0	-72.98	16.86	9.55	6.39	7.10
139	0	-7.12	8.94	0.57	0.38	0.43
139	0	-38.22	11.56	3.71	2.44	2.80
139	0	-68.09	16.09	8.68	5.69	6.55
144	0	-1.90	8.63	0.16	0.09	0.13
144	0	-33.33	11.45	3.43	2.02	2.77
148	0	-2.19	8.66	0.19	0.10	0.16
148	0	-31.23	11.61	3.49	1.85	2.96
150	0	-2.66	8.71	0.24	0.12	0.21
150	0	-30.58	11.77	3.62	1.81	3.13
151	0	-2.36	8.69	0.22	0.11	0.19
151	0	-27.97	11.52	3.32	1.61	2.91
156	0	-0.93	8.59	0.10	0.04	0.09
156	0	-24.51	11.71	3.42	1.39	3.12
160	0	-17.68	11.11	2.74	0.94	2.57
161	0	-16.44	11.03	2.64	0.86	2.50
166	0	-11.95	10.96	2.52	0.61	2.44
170	0	-8.98	11.11	2.64	0.46	2.60
174	0	-5.98	11.49	3.00	0.31	2.98
178	0	-1.40	10.40	1.90	0.07	1.90
184	0	2.42	10.09	1.60	-0.11	1.59
188	0	5.31	10.28	1.80	-0.25	1.78
190	0	6.94	10.39	1.91	-0.33	1.88